

TASK 05 — PROMPT EVALUATION & ACCURACY REPORT

VaultX Cyber Tech — Artificial Intelligence Track

1. Objective

The objective of this task was to evaluate the customer-support classification prompt using a small evaluation dataset, measure its **accuracy**, identify **classification errors**, **improve the prompt**, and compare the results before and after the prompt modification.

The classifier generates four structured fields:

- category
 - priority
 - sentiment
 - needs_human
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2. Evaluation Dataset

A dataset containing **10 realistic customer-support messages** was created.

Each sample contains:

- Customer message
- Expected classification
- Model prediction

The same dataset was used for evaluating the baseline and improved prompts.

3. Evaluation Metrics

Sample Accuracy

A sample is considered fully correct only when all four predicted fields match the expected output.

Accuracy = Fully Correct Samples / Successfully Evaluated Samples

Field Accuracy

Field accuracy measures how many individual fields were predicted correctly across the successfully evaluated samples.

The evaluated fields were:

- Category
 - Priority
 - Sentiment
 - Needs Human
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4. Baseline Evaluation

The initial prompt was evaluated before making any improvements.

Baseline Results

Metric	Result
Total samples	10
Successfully evaluated	7
API failures	3
Fully correct samples	3
Accuracy	42.86%
Field accuracy	85.71%

The three unsuccessful requests were caused by the Gemini API free-tier rate limit.

Therefore, the meaningful baseline accuracy among successfully evaluated samples was:

$$3 / 7 = 42.86\%$$

5. Baseline Error Analysis

The baseline evaluation identified the following errors:

Sample 1 — Category

Expected: billing

Predicted: refund

The customer reported being charged twice. Although a **refund** was requested, the primary issue was a **duplicate billing charge**.

Sample 2 — Sentiment

Expected: negative

Predicted: neutral

The customer reported being **unable to log into the account**, which was treated as a negative support issue in the expected classification.

Sample 3 — Priority

Expected: high

Predicted: medium

The package had been delayed for five days, but the model assigned medium priority.

Sample 6 — Human Escalation

Expected: false

Predicted: true

The refund request was treated as requiring human intervention even though the expected result was automatic handling.

6. Prompt Improvement

Based on the baseline errors, the classification prompt was improved.

The improved prompt introduced clearer rules for:

Category

The model was instructed to identify the **primary problem** instead of selecting a secondary requested action.

For example:

"I was charged twice and want a refund."

The primary category should be:

billing

rather than:

refund

Priority

Explicit definitions were added for:

- Critical
- High
- Medium
- Low

Sentiment

The prompt now distinguishes between:

- Positive
- Negative
- Neutral

Human Escalation

The prompt specifies that human escalation should only be selected for severe, urgent, unresolved, or manually actionable issues.

7. Second Evaluation

The improved prompt was evaluated using the same 10-sample dataset.

However, the Gemini API free-tier request-per-minute limit was reached after six successful requests.

Improved Prompt Results

Metric	Result
Total samples	10
Successfully evaluated	6
API failures	4
Fully correct samples	2
Accuracy	33.33%
Field accuracy	83.33%

The meaningful accuracy among successfully evaluated samples was:

$$2 / 6 = 33.33\%$$

The remaining four samples could not be evaluated because of the Gemini API 429 RESOURCE_EXHAUSTED rate-limit error.

8. Baseline vs Improved Prompt

Metric	Baseline	Improved
Total samples	10	10
Successfully evaluated	7	6
API failures	3	4
Fully correct	3	2
Accuracy	42.86%	33.33%
Field accuracy	85.71%	83.33%

9. Improvement Analysis

The second evaluation did **not** show an improvement in the measured accuracy.

Sample Accuracy

42.86% → 33.33%

Change:

−9.53 percentage points

Field Accuracy

85.71% → 83.33%

Change:

−2.38 percentage points

Therefore, the first prompt iteration did not improve the measured classification performance.

This result was recorded rather than artificially reporting an improvement.

10. Limitations

The evaluation was affected by the Gemini API free-tier request limit.

During the second evaluation, four out of ten samples returned:

429 RESOURCE_EXHAUSTED

because the API request-per-minute quota was exceeded.

Therefore, the second evaluation represents a **partial evaluation** and should not be considered a complete 10-sample comparison.

A larger and fully controlled evaluation should be performed when sufficient API quota is available.

11. Lessons Learned

The evaluation demonstrated several important prompt-engineering principles:

- Clear category definitions can reduce ambiguity.
 - Explicit priority rules help standardize classifications.
 - Primary-issue instructions are important when a message contains multiple concepts.
 - Structured output does not guarantee correct classification.
 - Prompt improvements must be validated using an evaluation dataset.
 - API rate limits can affect LLM evaluation experiments.
 - Evaluation results should be reported honestly, including unsuccessful API requests.
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12. Conclusion

The customer-support classification system successfully produced structured predictions and achieved strong field-level accuracy during the baseline evaluation.

The baseline achieved:

42.86% sample accuracy

85.71% field accuracy

After improving the classification prompt, the second evaluation achieved:

33.33% sample accuracy

83.33% field accuracy

The first prompt iteration therefore did not improve the measured results. However, the evaluation successfully identified specific classification weaknesses and demonstrated the complete prompt-evaluation workflow.

The API rate limit prevented all 10 samples from being evaluated during the second run, so further evaluation with sufficient API quota would be required for a more reliable comparison.